Phase 2B Mechanism-Diagnostic Change Report

12 November 2026 | verified development-only implementation and run

Purpose

We added audit-only diagnostics to determine whether honest agents encounter false-food pheromone. This work does not alter PPO inputs, simulator physics, policy parameters, or the attack threat boundary.

Implemented change	Verified boundary
False-food source component	Tracks applied mass through cap/evaporation; does not alter the actual field.
Normalized 3×3 audit views	Stored only in simulator metadata, never a PPO/detector input.
Exposure metrics	Counts honest exposed agent-steps and signal magnitude; corpus rejects leakage.
Replicated plan	Five predeclared action seeds; frozen policies/maps and delayed-200 attack retained.

Fresh replicated diagnostic

Three frozen Candidate J policies × four development maps × five stochastic action seeds produced 60 paired groups and 180 matched episodes. One of four agents was selected; delayed persistent injection began at transition 201 with a 200-unit cap.

Measure	Result
Mean injection loss	+0.17 honest food units
Median injection loss	0 units
Harmful groups	14 / 60 (23.3%)
Honest-exposure episodes	60 / 60
Mean honest exposed agent-steps	2,130.17
Applied false-food mass	200 units per attacked episode

Interpretation and verification

Injection loss is disabled honest delivery minus attacked honest delivery. The positive mean is small, the median is zero, and the result is mixed; it does not establish reliable disruption, a detector, a defense, or robustness. Focused tests: 76 passed. Full pinned suite: 164 passed in 32.86 seconds. Fresh artifacts are under artifacts/phase2b/mechanism-diagnostics-delayed-200-run-001/.

Next boundary

A different attack mechanism requires a separate accepted decision record. Final conclusions require unseen maps and a withheld attack configuration. Do not tune or select outcomes solely to increase mean loss.